

AURORA, OR, USA

WMO: 726959

Lat: 45.249N

Lon: 122.769W

Elev: 60

StdP: 100.61

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 94281

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.6	-2.1	-11.0	1.5	-0.7	-7.5	2.0	1.7	10.2	9.9	9.1	9.1	2.0	0	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	33.6	19.5	31.4	19.2	29.1	18.3	20.9	30.3	20.0	29.4	19.0	27.8	3.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.6	23.7	16.4	11.8	22.5	15.8	11.3	21.9	60.3	30.4	56.9	29.3	53.9	28.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.1	5.8	-6.5	37.8	2.5	2.1	-8.3	39.3	-9.8	40.5	-11.2	41.6	-13.0	43.1		
(5)			-7.1	22.5	2.6	1.2	-9.0	23.3	-10.5	24.1	-12.0	24.7	-13.9	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	5.4	6.6	8.6	10.7	14.3	17.1	20.6	20.7	17.7	12.4	8.0	5.0
	DBStd	6.35	3.40	2.86	2.76	3.09	3.46	3.30	2.95	2.87	3.11	2.84	3.32	3.58
	HDD10.0	579	146	98	62	26	3	0	0	0	0	10	76	159
	HDD18.3	2433	401	329	303	229	134	62	10	9	48	184	309	414
	CDD10.0	1417	3	3	17	47	137	212	327	333	232	85	16	4
	CDD18.3	228	0	0	0	1	9	24	79	83	30	1	0	0
	CDH23.3	2800	0	0	1	29	149	323	954	937	389	17	0	0
	CDH26.7	1118	0	0	0	5	44	122	407	398	139	2	0	0
Wind	WSAvg	2.2	2.4	2.5	2.5	2.3	2.1	2.2	2.1	1.9	1.9	2.0	2.3	2.4
Precipitation	PrecAvg	1069	153	96	127	80	62	41	9	12	45	95	154	180
	PrecMax	1415	339	172	229	160	123	84	28	47	183	258	370	374
	PrecMin	716	53	19	48	48	28	12	0	0	7	14	66	45
	PrecStd	183	78	40	47	28	29	18	7	12	37	60	69	82
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	16.4	21.3	27.1	31.1	34.6	37.0	36.3	33.6	26.1	18.0	15.9
		MCWB	12.3	9.8	13.0	16.4	18.6	20.0	20.8	20.1	18.9	15.3	14.7	12.7
	2%	DB	12.8	14.0	17.9	22.7	27.4	30.2	33.4	33.3	30.4	22.3	16.0	13.0
		MCWB	10.7	9.5	11.6	14.2	17.1	18.6	19.8	19.3	17.8	14.2	13.2	11.1
	5%	DB	11.5	12.4	15.7	19.2	24.1	27.2	31.1	31.1	27.8	19.6	14.1	11.4
		MCWB	9.7	9.1	10.6	12.3	15.9	17.6	19.5	19.3	17.5	13.7	11.8	10.1
	10%	DB	10.2	11.2	13.6	16.8	21.5	24.1	28.5	28.5	25.1	17.4	12.8	9.9
		MCWB	8.7	8.7	10.0	11.4	14.5	16.1	18.5	18.5	16.4	13.2	10.8	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	12.3	14.2	17.1	20.0	21.2	22.9	21.9	20.7	17.7	16.2	13.4
		MCDB	14.5	13.6	18.9	24.7	28.7	31.4	33.1	33.1	29.6	21.2	17.1	15.1
	2%	WB	11.5	11.0	12.6	14.9	18.0	19.3	20.9	20.8	19.2	15.8	13.9	11.8
		MCDB	12.3	12.6	15.7	21.3	25.1	28.6	29.7	30.4	27.6	19.5	15.4	12.7
	5%	WB	10.0	10.0	11.5	13.2	16.5	18.0	19.8	19.8	18.1	14.7	12.4	10.3
		MCDB	11.1	11.7	14.4	18.1	22.9	25.9	29.7	29.2	25.4	17.8	13.6	11.4
	10%	WB	8.8	8.8	10.5	11.8	15.1	16.7	18.8	18.8	17.0	13.9	11.2	8.6
		MCDB	10.0	10.7	13.2	15.6	20.8	23.2	28.0	27.4	23.5	16.7	12.5	9.5
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.4	8.5	9.7	10.6	11.2	14.0	14.0	12.8	9.4	6.9	5.7
		MCDBR	7.0	9.6	12.8	15.6	16.5	16.7	18.3	18.3	18.0	14.2	7.9	6.5
		MCWBR	5.0	6.0	7.2	8.0	8.0	7.4	7.0	6.9	7.4	7.3	5.2	5.0
	5% WB	MCDBR	6.2	7.4	10.4	13.4	14.3	15.1	16.9	16.8	15.5	10.3	6.4	6.0
		MCWBR	5.0	5.2	6.5	7.4	7.4	7.1	7.2	6.9	7.0	6.1	4.7	5.0
Clear-Sky Solar Irradiance	taub	0.317	0.315	0.324	0.353	0.368	0.356	0.346	0.351	0.339	0.332	0.317	0.320	
	taud	2.538	2.554	2.535	2.419	2.394	2.441	2.493	2.497	2.511	2.542	2.549	2.505	
	Ebn at Noon	801	878	915	908	900	910	915	900	884	843	794	753	
	Edh at Noon	66	77	89	109	115	111	104	100	90	77	64	62	
All-Sky Solar Radiation	RadAvg	1.20	2.03	2.95	4.29	5.47	6.08	6.94	5.93	4.42	2.57	1.39	0.98	
	RadStd	0.20	0.29	0.39	0.40	0.54	0.56	0.47	0.39	0.38	0.31	0.19	0.15	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+48
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	+26

Nomenclature: See separate page